Example 2-1

H_0: Data follow a normal distribution.
H_1: Data do not follow a normal distribution.

$\hat{\mu} = \bar{X} = 89.871$
$\hat{\sigma}^2 = S^2 = 52.531^2$

cell	f_i	P_i	$e_i = np_i$	$\frac{(f_i - e_i)^2}{e_i}$
[0, 25)	7	$P(0 \le X < 25) = P(\frac{0-89.871}{52.531} < Z < \frac{25-89.871}{52.531}) = 0.0646$	$132(0.0646) = 8.531$	$\frac{(7-8.531)^2}{8.531} = 0.2747$
[25,50)	26	$P(25 \le X < 50) = 0.1155$	15.246	7.5855
[50,75)	29	$P(50 \le X < 75) = 0.1650$	21.785	2.3895
[75,100)	25	0.1885	24.732	0.00296
[100,125)	17	0.1722	22.729	1.44403
[125,150)	9	0.1257	16.597	3.47774
[150,175)	8	0.0734	9.689	0.29442
[175,200)	3	0.0343	4.522	2.7394
[200,225)	6	0.0128	1.687	
[225,250)	2	0.0038	0.503	
Total	132			18.208047

Test statistic:
$\chi^2 = \sum_{i=1}^{8} \frac{(f_i - e_i)^2}{e_i}$
$= 18.208047$

$df = k - d - 1$
$= 8 - 2 - 1 = 5$

p-value
$= P(X > 18.208047)$
$= 0.0027 < \alpha = 0.05$
where $X \sim \chi^2(5)$.

Conclusion:
We reject H_0 and conclude that data do not follow a normal distribution at $\alpha = 0.05$.

Example 2.1 (9)

H_0: The frequency follows a Poisson distribution.
H_1: The frequency does not follows a Poisson distribution.

$$\hat{\lambda} = \bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{173}{98} = 1.765.$$

x_i	f_i	P_i	$e_i = nP_i$	$\frac{(f_i - e_i)^2}{e_i}$
0	19	$P(X=0) = \frac{e^{-\hat{\lambda}}\hat{\lambda}^0}{0!} = 0.1712$	$98(0.1712) = 16.78$	$\frac{(19-16.78)^2}{16.78} = 0.295$
1	26	$P(X=1) = \frac{e^{-\hat{\lambda}}\hat{\lambda}^1}{1!} = 0.3021$	29.61	0.441
2	27	0.2667	26.13	0.029
3	13	0.1569	15.37	0.367
4	11	0.0692	6.78	0.826
≥ 5	2 (13)	$P(X \geq 5) =$ 0.0339	3.33 (10.11)	
Total	98			1.964

Test statistic:

$$\chi^2 = \sum_{i=1}^{5} \frac{(f_i - e_i)^2}{e_i} = 1.964$$

$df = k - d - 1$
$= 5 - 1 - 1 = 3$

p-value $= P(Y > 1.964)$ where $Y \sim \chi^2_{(3)}$
$= 0.5799 > \alpha = 0.05$.

not enough evidence to reject H_0. So,

$\therefore$ H_0 is not rejected and we can conclude that the frequency follows a Poisson distribution at $\alpha = 0.05$.